

Shubhanshu Tiwari

Noida, Uttar Pradesh, India

+91-9161271367

shubhanshutiwari609@gmail.com

LinkedIn

GitHub

PROFILE

AI/ML Engineer and Computer Science student with hands-on experience building end-to-end intelligent systems spanning computer vision, NLP, and generative AI. Focused on delivering production-ready applications that bridge deep learning research with practical deployment using Python, Streamlit, and modern LLM orchestration frameworks.

EDUCATION

Galgotias University

Greater Noida, India

Bachelor of Computer Applications – Computer Programming & AI/ML

Aug 2024 – Mar 2027

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, Python, Statistics, DBMS, Computer Systems Networking

CERTIFICATIONS

AWS APAC – Solutions Architecture Job Simulation • Cisco Network Certifications • Linux Unhatched • Infosys Springboard Virtual Internship Certificate • Oracle Database 11g Performance Tuning

TECHNICAL PROJECTS

Medical AI: Brain Tumor Detection & RAG Clinical Report System | Python, TensorFlow, LangChain, ChromaDB, Gemini

- Architected an end-to-end medical decision support system combining CNN-based transfer learning to classify brain MRI scans into 4 categories: Glioma, Meningioma, Pituitary Tumor, and Healthy.
- Built a multimodal RAG pipeline using ChromaDB vector store with `all-MiniLM-L6-v2` HuggingFace embeddings to retrieve context-matched clinical guidelines, piped into Gemini-2.5-Flash for deterministic, hallucination-minimized report generation.
- Integrated visual anomaly contrast rendering – auto-generating side-by-side patient scan vs. healthy baseline comparisons for explicit clinical auditing.
- Deployed as a production-ready, cloud-deployable Streamlit dashboard; performance tracked via Loss Convergence Curves and Categorical Accuracy Trends across stratified validation sets.

SentiScope – NLP Sentiment Analyzer | Python, Scikit-learn, TF-IDF, NLTK, Streamlit | GitHub

- Built a real-time sentiment classification web app using a Logistic Regression classifier with TF-IDF vectorization (10,000 features, unigram+bigram, sublinear scaling) trained on the NLTK Movie Reviews corpus plus 60 hand-crafted examples.
- Engineered confidence scoring using multi-class probability outputs with session-persistent history tracking, sentiment analytics dashboard, and detailed class probability breakdowns.
- Implemented `@st.cache_resource` model caching to eliminate re-training on cold starts; preprocessing pipeline handles URL removal, lowercasing, and special character filtering.

Brain Tumor Detection – ML Classifier | Python, Deep Learning, Streamlit | GitHub

- Developed a standalone CNN-based MRI classification system with a pre-trained `brain_tumor_model.h5` weight file, exposed via an interactive Streamlit interface for rapid inference on uploaded scans.

StreamliteML | Python, Streamlit, Machine Learning | GitHub

- Interactive ML demonstration platform built with Streamlit, providing hands-on model exploration and visualization for supervised learning workflows.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, C, C++ Ooops, HTML/CSS,Sql

AI & ML Frameworks: LangChain, TensorFlow, Scikit-learn, PyTorch, ChromaDB, NLTK, HuggingFace

ML Algorithms: Logistic Regression, Decision Trees, Random Forest, K-Nearest Neighbours, SVM, CNN, Transfer Learning

Core Competencies: RAG Architectures, NLP, Computer Vision, Sentiment Analysis, Medical Imaging, Deep Learning

Tools & Platforms: Streamlit,Replit,Vs Code,Antigravity,Collab, Git, GitHub, Jupyter Notebook, Google Generative AI (Gemini)